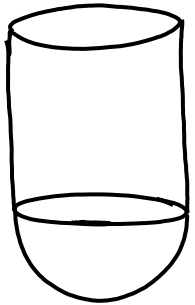


(1) **The radius of base of a solid cylinder is 7 cm and its height is 21 cm. It is melted and converted into small bullets. Each bullets is of same size. Each bullet is consisted of two parts viz. a cylinder and a hemisphere on one of its base. The total height of bullet is 3.5cm an radius of base is 2.1 cm. Approximately how many complete bullets can be obtained?**

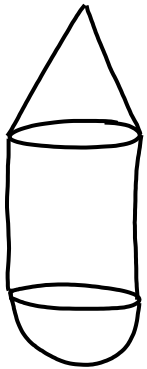
a) 83 **b) 89**
c) 84 **d) 79**



② **The lower and upper part of a water tank is hemispherical and cylindrical, respectively. If its radius is 12 m and volume is $3312\,\pi$ cubic m then find the height of the cylindrical part.**

a) 12 **b) 13**
c) 14 **d) 15**

D.M.R.C.(J.S.C) 2013



(3) **A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm. and the total height of the structure is 12.8 cm. The volume V of the structure lies between**

a) $370\ cm^3$ and $380\ cm^3$
b) $380\ cm^3$ and $390\ cm^3$
c) $390\ cm^3$ and $400\ cm^3$
d) None of these

CDS 2014 I

4 The ratio of curved surface area of a right circular cylinder to the total area of its two bases is 2:1. If the total surface area of cylinder is 23100cm^2, then what is the volume of cylinder?

a) 247200 b) 269500
c) 312500 d) 341800

$V = 4312$

$\Rightarrow \frac{22}{7} \times a^2 \times 4a = 4312$

(5) (HW) A right circular cylinder is formed. A= sum of total surface area and the area of the two bases. B = the curved surface area of this cylinder. If A : B = 3 :2 and the volume of cylinder is $4312cm^3$, then what is the sum of area (in cm^2) of the two bases of this cylinder?

a) 154 b) 308
c) 462 d) 616

⑥ **The ratio of curved surface area and volume of a cylinder is 1:7. The ratio of total surface and volume is 187 : 770. What is the respective ratio of its base radius and height?**

a) 5 : 8 **b) 4 : 9**
c) 3 : 7 **d) 7 : 10**

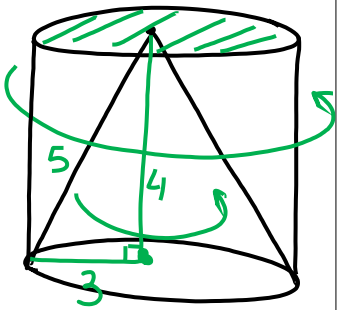
⑦ **The radii of two cylinders in ratio 3:5 and their heights are in ratio of 2:3. Find the ratio of their curved surface areas.**

a) 5:2 **b) 6:25**

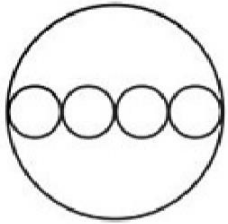
c) 2:5 **d) 25:6**

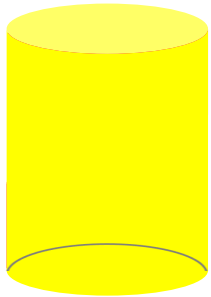
(8) From a solid right circular cylinder of length $4\ cm$ and diameter $6\ cm$, a conical cavity of the same height and base is hollowed out. The whole surface of the remaining solid is

a) $48\ \pi$ b) $15\ \pi$
c) $63\ \pi$ d) 24



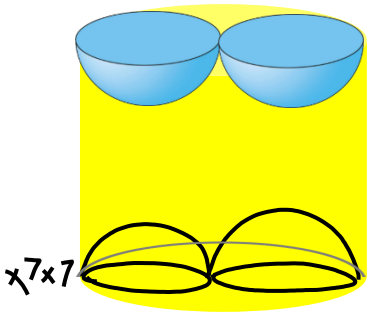
CGL 2014 T-2





(9) **A solid cylinder has radius of base 14cm and height 15cm. 4 identical cylinders are cut from each base as shown in the given figure. Height of small cylinder is 5cm what is the total surface area of the remaining part?**

a) 3740 **b) 3432**
c) 3124 **d) 2816**



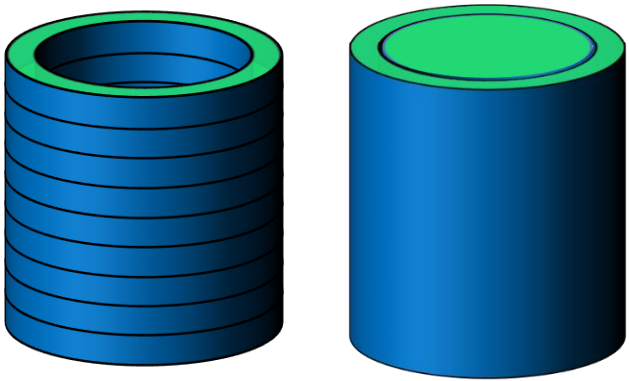
(10) **A right circular cylinder has height 28cm and radius of base 14cm. Two hemisphere of radius 7cm each are cut from each of the two bases of the cylinder. What is the total surface area of the remaining part?**

a) 3842 **b) 4312**
c) 3296 **d) 4436**

CGL 2017 T-2

(11) If $1\ cm^3$ of cast iron weights 21 gms, then the weight of a cast iron pipe of length 1 m with a bore of 3 cm and in which the thickness of the metal is 1 cm, is:

a) 46.2 kg b) 24.2 kg
c) 26.4 kg d) 18.6 kg

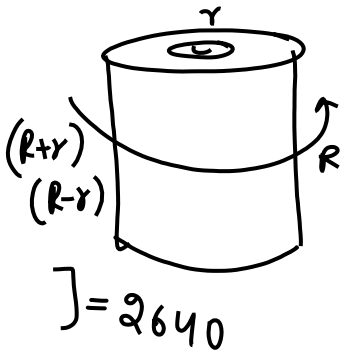


(12) **A well with 14 m inside diameter is dug 10 m deep. Earth taken out of it, has been evenly spread all around it to a width of 21 m to form an embankment. The height of embankment (in meters) is.**

a) .33 m **b) .66 m**
c) 1.33 m **d) 1.66 m**

(13) A hollow cylinder is made up of metal. The difference between outer and inner curved surface area of this cylinder is 352 cm^2. Height of the cylinder is 28cm. If the total surface area of this hollow cylinder is 2640cm^2, then what are the inner and outer radius (in cm)?

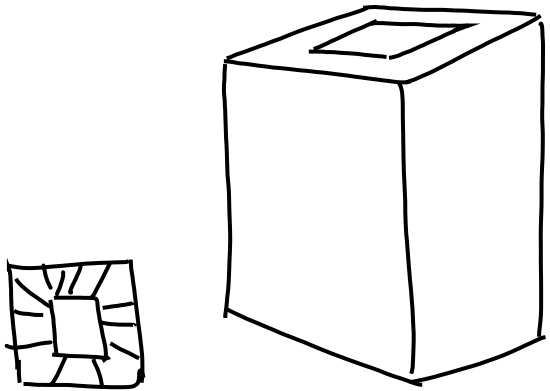
a) 4, 6 b) 10, 12
c) 8, 10 d) 6, 8

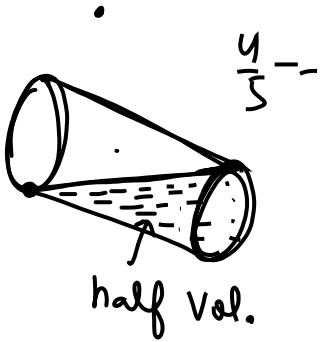


(14) **A hollow square shaped tube open at both ends is made of iron. The internal square is of 5 cm side and the length of the tube is 8 cm. There is 192 cm^3 of iron in the tube. Find its thickness:**

a) 2 cm **b) 7 cm**

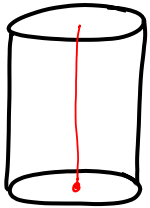
c) 1 cm **d)Can't determined**





15 A cylinder is filled to $\frac{4}{5}th$ of volume. It is then tilted so that the level of water touches a point on the circumference of top and diagonally opposite point on the circumference of bottom. In the process, 30 litre of the water is spilled. What is the volume of the cylinder?

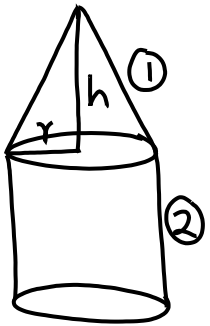
a) 75 litre
b) 96 litre
c) Data insufficient
d) 100 litre



16 A right circular cylinder having diameter $21\ cm$ and height $38\ cm$ is full of ice cream. The ice cream is to be filled in cones of height 12 and diameter $7\ cm$ having a hemispherical shape on the top. The number of such cones to be filled with ice cream is

a) 54 b) 44
c) 36 d) 24

(17) A solid consists of a circular cylinder with exact fitting right circular cone placed on the top. The height of the cone is h. If total volume of the solid is three times the volume of the cone, then the height of the circular cylinder is



a) 4 h

b) $\frac{3h}{2}$

c) 2h

d) $\frac{2h}{}$

3

CGL 2012 T-2



18
HW

A cylinder and a cone have equal radii of their bases and equal heights. If their curved surface areas are in the ratio $8:5$, the ratio of their radius and height is

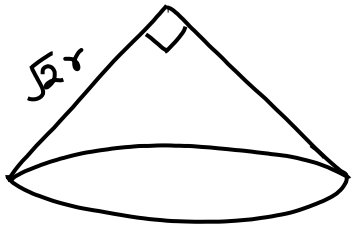
a) $1:2$ b) $1:3$
c) $2:3$ d) $3:4$

CGL 2011 T-2

(19) The vertical angle of a right circular cone is $\frac{\pi}{2}$ and the slant height is $\sqrt{2}$ r cm. What is the volume of the cone in cubic cm?

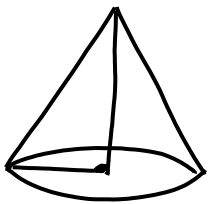
a) πr b) $9\pi r$

c) $\frac{\pi r^3}{3}$ d) $3\pi r^3$



CDS 2017 II

20

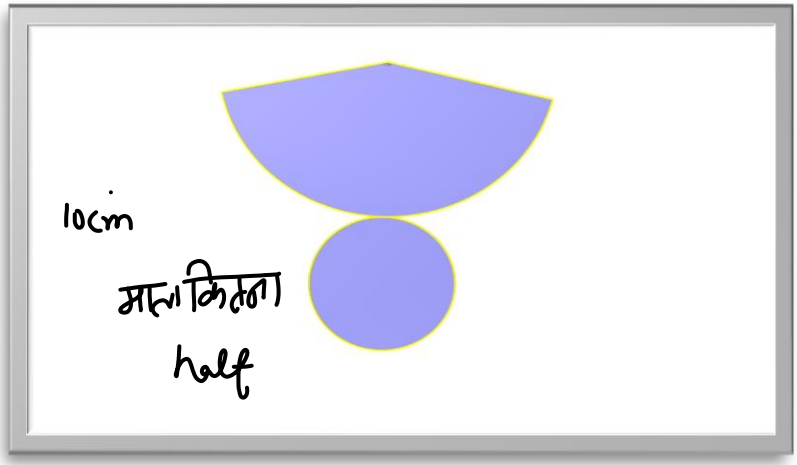


If h, c, v are respectively the height, curved surface area and volume of a right circular cone, then the value of $3\ \pi v\ h^3 - c^2h^2 + 9v^2$ is

a) 2 b) – 1

c) 1 d) 0

CGL 2013 T-2

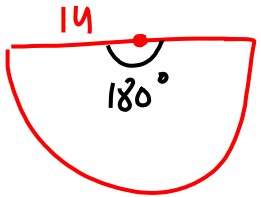


(21) **A cardboard sheet in the form of a circular sector of radius 30 cm and central angle 144° is folded to make a cone. What is the radius of the cone?**

a) 12 cm b) 18 cm
c) 21 cm d)none of these

CDS 2012 I

22 A semi-circular thin sheet of a metal of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup?



a) $\frac{1000}{3}\sqrt{3}\ cm^3$

b) $300\sqrt{3}\ cm^3$

c) $\frac{700}{3}\sqrt{3}\ cm^3$

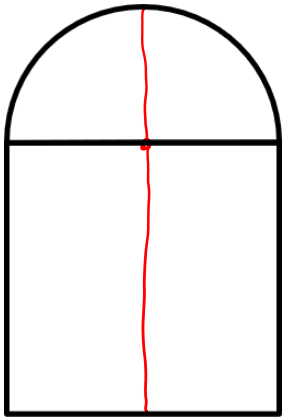
d) $\frac{1078}{3}\sqrt{3}\ cm^3$



23 From a circular sheet of paper of radius $10\ cm$, a sector of area 40% is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and the height of the cone is

a) $3:4$ b) $4:3$
c) $1:2$ d) $1:1$

CGL 2012 T-2



24 HW

A hemisphere is kept on the top of a cube. Its front view Is shown in the given figure. The total height of the figure is 21 cm. The ratio of curved surface area of the hemisphere and total surface area of cube is 11:42. What is the total volume (in cm^3) of figure?

a) 3318.33 b) 3462.67

c) 3154.67 d) 3248.33

CGL 2017 T-2

25

A solid sphere has a radius $12cm$. It is melted to form a cube. 20% material is wasted in this process. The cube is melted to form hemisphere. In this process $16.\ 66\%$ material is wasted. The hemisphere is melted to form two spheres with radii in the ratio $1:3.\ 12.\ 5\%$ material was wasted in this process. What is the radius of the new bigger sphere?

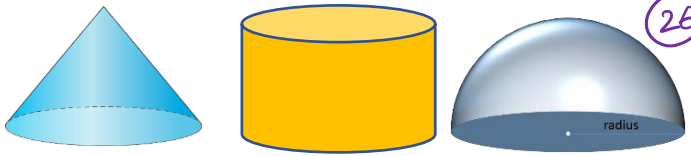
a) $\sqrt[3]{36}$

b) 6

c) 18

d) $3\sqrt[3]{36}$

Based on CGL 2017 T-2



26

A cone, a cylinder and a hemisphere stand on equal bases and have equal heights. The ratio of their volumes is

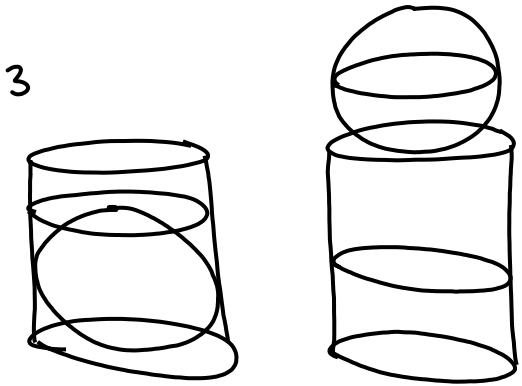
a) 2:3:1 **b) 2:1:3**

c) 1:3:2 **d) 1:2:3**

(27) **A cylindrical can whose base is horizontal and is of internal radius 3.5 cm contains sufficient water so the when a solid sphere is placed inside, water just covers the sphere, the sphere fits in the can exactly. The depth of water in the can before the sphere was put is:**

a) $\frac{35}{3}\ cm$ b) $\frac{17}{3}$ cm

c) $\frac{7}{3}$ cm d) $\frac{14}{3}$ cm





28 The internal and external radii of a hollow hemispherical vessel are $6\ cm$ and $7\ cm$ respectively. What is the total surface area (in $c\ m^2$) of the vessel?

a) 183π b) 189π
c) 177π d) 174π

CGL 2018 T-2



24 A metallic hemisphere bowl is made up of steel . The total steel used in making the bowl is $342\pi\ cm^3$. The bowl can hold $144\pi\ cm^3$ water. What is the thickness (in cm) of bowl and the curved surface area of outer side?

a) 6, 162 π b) 3, 162 π
c) 6, 81 π d) 3, 81 π

CGL 2017 T-2